

Math 322: Commutative Algebra

Chapter 1: Euclidean Domains and Principal Ideal Domains

- **Ring** Abelian group $(R, +)$ equipped with a multiplication that satisfies left and right distributivity property and contains a multiplicative identity
- **Commutative Ring** If the multiplication is commutative
- **Zero Divisor** Let R be a commutative ring. An element $a \in R \setminus \{0\}$ is a non-zero divisor if there exists $b \in R \setminus \{0\}$ such that $ab = 0$.
- **Integral Domain** A commutative ring with no zero divisors, written ID.
- *Property of ID:* If $0 \neq a \in R$ and $b, c \in R$ such that $ab = ac$ then $b = c$.
- **Ideal** A subgroup I of $(R, +)$ such that $ab, ba \in I$ for any $a \in R$ and $b \in I$.
- **Generating Set** A generating set for an ideal I is a subset S of I , such that the elements of S R -generate I .
- **Ideal of R generated by S** If $S = \{s_1, s_2, \dots, s_n\}$ is finite then $I = s_1R + s_2R + \dots + s_nR$ is the ideal of R generated by S .
- **Principal Ideal** An ideal I is principal if I can be generated by single element.
- Given two ideals I, J in a commutative ring R , we can construct new ideals:

$$\begin{aligned} I \cap J, \\ I + J, \\ IJ = \left\{ \sum_{\text{finite}} ab : a \in I, b \in J \right\} \end{aligned}$$

- *Notation.* We write $a \equiv b \pmod{I}$ for $a - b \in I$.
- **Principal Ideal Domain** An integral domain such that any ideal is principal.

- Main instances of PID's:
 1. Ring of Integers
 2. Polynomial rings in one variable over a field
- **Euclidean Function** Let R be an integral domain. A Euclidean Function on R is a function $v : R \setminus \{0\} \rightarrow \mathbb{N}_0$ such that the following hold:
 - (i) $v(ab) \geq v(a)$ for all $a, b \in R \setminus \{0\}$
 - (ii) for any $a \in R$ and any $b \in R \setminus \{0\}$, there exists $q, r \in R$ such that $a = qb + r$ and either $r = 0$ or $v(r) < v(b)$
- **Euclidean Domain** ID R such that there is a Euclidean Function on R .
- $\text{ED} \implies \text{PID}$
- *Important theorem relating a ring and its polynomial ring* Let R be a ring and $R[x]$ the polynomial ring over R . TFAE
 - (i) $R[x]$ is a ED
 - (ii) $R[x]$ is a PID
 - (iii) R is a field